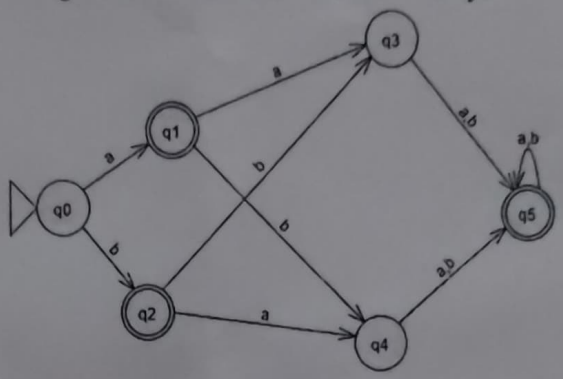
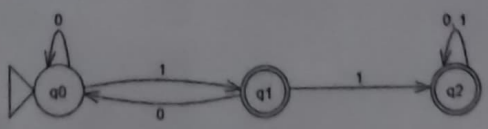
B.Tech. Comp. Engg. Vth Sem Examination 2022
AUTOMATA THEORY
Paper no CEN-502

Max Time : Three hours | Max. Marks: 60

Attempt any TWO parts from each question.

Write your Roll No here 20BCS021 *on receipt of the question paper immediately*

Q. No	CO	Questions	Marks
Q1 a)	1	State the mathematical description of a Finite Automaton. Also, Differentiate between a Transition Diagram and DFA using suitable examples.	6
Q1 b)	1	Construct a DFA for the following and validate your answers: i) ((1+0)*0) + 0)* ii) (0(0+00)* + 1)*	6
Q1 c)	1	Minimize the following DFA ***with the Matrix Method only***.  **Verify your answers using suitable sentences.**	6
Q2 a)	2	Construct a regular expression for the following languages $\Sigma = \{0,1\}$ i. Set of all strings with even 0s ii. Set of all strings with the number of 1s being in multiples of 3 **Validate your answer with suitable reasons.**	6
Q2 b)	2	Mathematically, state and prove the Arden's theorem. Also, explain the usefulness/application of this theorem.	6
Q2 c)	2	Construct the Regular Expression for the given Automaton showing all intermediate steps.  **Verify your answers using suitable sentences.**	6

Q3 a)	2	Explain the concept of Ambiguity in Grammars using suitable examples. Also explain the Chomsky classification of Languages in details.	6
Q3 b)	3	Construct a CFG for a language described as follows : $L_1 = \{ 1^*0^* + 0^n1^n$, **where** $n > 1\}$ **Verify your answers using suitable sentences.**	6
Q3 c)	3	Simplify the following CFG. $Z \rightarrow AZC \mid aAb$ $B \rightarrow bB \mid ZC$ $J \rightarrow ABt$ $A \rightarrow aC \mid CB$ $C \rightarrow d \mid \varepsilon$ **Mention all the steps and the corresponding rules used at appropriate positions.**	6
Q4 a)	4	Mathematically define a PDA. Also construct a PDA for a Language that accepts a set of all strings over {a, b} which contain equal no 'a' & 'b'. **Validate your answers with suitable examples.**	6
Q4 b)	4	Construct a Deterministic PDA for the followings: i) Accepting set of all strings over {0, 1} which do not contain 1001. ii) $L = \{ w_1cw_2$: such that $\lvert w_1 \rvert > \lvert w_2 \rvert$ and $w \varepsilon \{a,b\}^*\}$. **Validate your answers using suitable examples**	6
Q4 c)	4	Mention the steps to convert a CFG to a PDA along with the advantages as well as disadvantages of this process. Also convert the following CFG to PDA: $Z \rightarrow AZC \mid aAb$ $B \rightarrow bB \mid ZC \mid \varepsilon$ $J \rightarrow ABt$ $A \rightarrow aC \mid CB$ $C \rightarrow d \mid \varepsilon$	6
Q5 a)	5	Construct a Turing Machine for the followings operations: i) $f(n, m) = n*m$; ii) $\sqcup w \sqcup \rightarrow \sqcup w^r \sqcup$ [*Where* w^r *stands for reverse of string* w.] **Verify your answer with suitable examples.**	6
Q5 b)	5	What are primitive recursive functions? Show that the $f(x, y) = x^2y + 2y^3$ is primitive recursive.	6
Q5 c)	5	Explain the following using suitable examples: i) Recursive Language Vs Recursively Enumerable Languages ii) P problems vs. NP Hard Problems	6

**********End of Paper**********